UNDER REVIEW

RESCIENCE C

Replication / Computational Neuroscience

[Re] Dreaming neural networks: forgetting spurious memories and reinforcing pure ones

Clément Saillant[1, ID]

[1]Hypneum Lab (L'Electron Rare), Grandris, France

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1 Introduction

Fachechi, Agliari and Barra[1] (Neural Networks 112, 2019, 24–40; arXiv:1810.12217) revisit the "unlearning" idea of Hopfield, Feinstein and Palmer[2] and derive a closed-form "dreaming" kernel for the Hopfield model. Starting from the Hebbian coupling matrix $J = \frac{1}{N}\sum_{\mu=1}^{P} \xi^{\mu}\xi^{\mu\intercal}$ built from $P = \alpha N$ random ± 1 patterns, a sleep phase of duration t produces

$$J(t) = J\,(1+t)\,(\mathbb{1} + tJ)^{-1},$$

which equals J at $t = 0$ and tends to the projection (pseudo-inverse) rule of Kanter and Sompolinsky[3] as $t \to \infty$. The article's statistical-mechanics analysis predicts that the critical storage load $\alpha_c(t)$ rises from the Amit–Gutfreund–Sompolinsky value[4] ≈ 0.138 at $t = 0$ towards the theoretical bound $\alpha = 1$, that spurious (mixture) attractors are removed, that pure patterns are reinforced, and that the gain is robust to thermal noise. The original Monte Carlo checks (Section 2.5 of the published article) use, for the magnetisation curves, single-spin-flip Glauber dynamics measured after equilibration on networks of $N = 5\,000$ neurons averaged over $M = 10$ pattern realisations (their Fig. 7, loads $P/N \in \{0.02, 0.08, 0.32\}$, $t \in \{0, 0.1, 1, 10, 100, 1000\}$) and a finite-size comparison at $N \in \{500, 1\,000, 5\,000\}$ (their Fig. 8). The zero-noise internal-field and basin analyses (their Figs. 9–10) use the *parallel* sign update of their Eq. (2.17) at $N = 1\,000$, $P = 50$. For the fields (Fig. 9) the network starts from random configurations, 20 evolutions for each of 20 pattern realisations; for the basins (Fig. 10) it starts from the first stored pattern with each bit flipped independently with probability p, success being declared when that pattern carries the largest Mattis overlap, 200 evolutions for each of 200 realisations. The replica-symmetric capacity rises from $\alpha_c(0) \simeq 0.138$ to $\alpha_c \simeq 1.07$ as $t \to \infty$, the excess over Gardner's bound being attributed to replica-symmetry breaking. No capacity curve is measured by Monte Carlo in the original: the simulations verify magnetisations, fields and basins against the theory.

We independently implement the published kernel and *partially* replicate its qualitative claims at finite size ($N \leq 1\,000$) with a new, noisy-cue capacity estimator and new random seeds: we test the retrieval gain, the removal of spurious final states, the robustness to thermal noise and the projector limit, and we do not reproduce the replica-symmetric capacity curve nor the original Monte Carlo figures. Each of the three studies was pre-registered and frozen by an annotated git tag before its own main run; the later registrations were informed by the preceding studies (Section "Pre-registration").

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Correspondence should be addressed to Clément Saillant (clement@saillant.cc)
The authors have declared that no competing interests exist.
Code is available at https://github.com/hypneum-lab/rescience-nh01..
Data is available at https://github.com/hypneum-lab/rescience-nh01/tree/main/data – DOI 10.5281/zenodo.23833431.

The replication was carried out inside the Hypneum Lab research programme (issue hypneum-lab/nerve-wml#68), where the dreaming kernel serves as a reference operation; the replication stands on its own and makes no claim about that programme.
Beyond replicating, we add one control absent from the original work: a uniform multiplicative rescaling ("downscale") of the Hebbian couplings, matched to the dreamed kernel on the Frobenius norm. This control answers a question the original article does not ask — whether the benefit of dreaming could be reproduced by a global reduction of synaptic strength, as in homeostatic-downscaling accounts of sleep[5] — and it provides a calibrated null for the capacity ratio. Finally, we test the kernel on correlated patterns, which the original article treats only for independent patterns.

2 Methods

2.1 Original model

Binary units $s_i \in \{\pm 1\}$, $i = 1 \dots N$; patterns $\xi^\mu \in \{\pm 1\}^N$ i.i.d. uniform; Hebbian kernel as above; dreamed kernel $J(t)$. In pattern space, $J_{ij}(t) = \frac{1}{N} \sum_{\mu\nu} \xi_i^\mu [(1+t)(\mathbb{1} + tC)^{-1}]_{\mu\nu} \xi_j^\nu$ with $C_{\mu\nu} = \frac{1}{N} \xi^\mu \cdot \xi^\nu$. The eigenvalues $\lambda \geq 0$ of J map to $\lambda(1+t)/(1+t\lambda)$. Retrieval is assessed by the overlap $m = \frac{1}{N} \sum_i s_i \xi_i^\mu$ of the final state with a stored pattern.

2.2 Implementation

Our code (S1: `run.py`, `confirmatory.py`, `correlated.py`; 713 + 384 + 237 lines, NumPy only, no original code reused) computes $J(t)$ by spectral decomposition of the symmetric J (`numpy.linalg.eigh`, negative numerical eigenvalues clipped to 0). Twenty test functions (23 executed cases) check that $J(0) = J$, that the spectral form equals the pattern-space closed form on 2-pattern and many-pattern cases, that $t = 10^8$ approximates the projector within the tested tolerance, that the matched shrink is exactly inert at zero temperature (S2), and cover the confirmatory and correlated runners. The diagonal J_{ii} is set to 0 in every arm before the dynamics. The dynamics are asynchronous with one fresh random permutation of the N units per sweep, shared by the probes of a cell. At $T = 0$, $s_i \leftarrow \mathrm{sign}(h_i)$ with ties ($|h_i| \leq 10^{-12} N \max|J|$) resolved by keeping the current spin — a tolerance chosen so that the rule is exactly invariant under $J \to cJ$; the run stops at the first sweep without a flip, cap 100 sweeps (non-convergence counted as not retrieved; none occurred in any run). At $T = 0.2$, Glauber updates with $P(s_i = +1) = 1/(1 + e^{-2\beta h_i})$, $\beta = 5$, 40 sweeps, overlap averaged over the last 10.
Cue: a stored pattern with exactly round$(0.10\,N)$ bits flipped (overlap 0.80 at $N = 1\,000$; 0.8125, 0.797, 0.797 at $N = 64$, 128, 256 because of rounding). Probes: the first $\min(P, 32)$ stored patterns, i.e. 10 and 20 probes at the loads 0.01 and 0.02 of the correlated run. The pilot also includes a descriptive $T = 0$ condition with $f = 0$ (unperturbed stored-pattern cues), absent from the two $N = 1\,000$ studies. Final states are classified as `retrieved` ($m \geq 0.9$; 0.8 at $T = 0.2$), `other_stored` or `spurious`. Capacity estimator per (arm, seed): $\hat{\alpha}_c$ is the linearly interpolated 0.5 crossing of the retrieval fraction $r(\alpha)$ immediately after the *last* grid load with $r \geq 0.5$, on the grid $\alpha \in \{0.04, 0.06, \dots, 0.20, 0.25, \dots, 0.95\}$ (24 values; 0.01 and 0.02 added for the correlated run). If the largest load still has $r \geq 0.5$ the estimate is right-censored at 0.95; if no load reaches 0.5 the code records the flag `below_grid` and a zero sentinel, which marks censoring, not a measured zero capacity. Patterns are nested in α and the same patterns, cues and update orders are used for every arm of a cell (paired design). Statistics are paired per-seed ratios and differences, mean over 10 seeds with a 95 % percentile bootstrap CI (10 000 seed-level resamples, bootstrap seed 20261025). Every random draw comes from `numpy.random.default_rng([20261025, seed, N, stream])`.

2.3 Matched uniform-shrink control (added)

For each $t > 0$, $J_c = c(t)\,J$ with $c(t) = \|\mathrm{offdiag}\,J(t)\|_F / \|\mathrm{offdiag}\,J\|_F$, i.e. equal global synaptic budget; c depends on t, α, N and the seed and the correlation level; across the shrink arms it ranges from 0.0884 to 0.9985. Two facts were declared before any run so that the control cannot be misread: at $T = 0$ the sign dynamics is invariant under $c > 0$, so the control must give a ratio of exactly 1 (an implementation check, not an empirical hypothesis); at $T > 0$, J_c at inverse temperature β has the same transition probabilities as J at $c\beta$, i.e. the control is the Hebbian network at temperature T/c. (The same rescaling of β makes the $(1+t)$-normalised kernel used here and its unnormalised ridge form statistically equivalent: their partition functions coincide up to $\beta \to \beta(P/N)(1+t)/t$, as noted by Agliari et al.[6] in their footnote 9 and in footnote 2, p. 117, of Aquaro's thesis[7]; at $T = 0$ the normalisation has no effect.) This identity does not impose a deterministic inequality on a finite-time, thresholded, single-realisation capacity estimate (in the pilot, seed 6 at $N = 256$, $T = 0.2$ gives a shrink ratio of 1.012), so the effect at $T = 0.2$ is measured empirically.

2.4 Correlated patterns (added)

One ancestor $a \in \{\pm 1\}^N$ per (seed, level); each pattern bit equals a_i with probability $(1 + m)/2$, so the mean pairwise overlap is m^2. Levels $m \in \{0, 0.15, 0.30\}$ ($q \in \{0, 0.0225, 0.09\}$), seeds 100–109 (disjoint from the independent-pattern seeds 0–9).

2.5 Pre-registration and decision rules

Three L1 pre-registrations, each frozen by an annotated tag pushed to GitHub before launch and each archived by Software Heritage (verbatim copies in `prereg/` of the submission repository, S3):

- **Pilot** (tag of 2026-10-25, S3): $N \in \{64, 128, 256\}$, $t \in \{0, 0.3, 1, 3, 10, 30, 100\}$, three conditions.
- **Confirmatory** (second tag of 2026-10-25): $N = 1\,000$, $t \in \{0, 1, 3, 5, 7, 10, 15, 20, 30, 100\}$, $T \in \{0, 0.2\}$.
- **Correlated** (tag of 2026-10-27): as confirmatory, three correlation levels, 26 loads.

Binding criteria at $t = 10$: **C1** (gain) confirmed if the CI lower bound of the ratio R of the dreamed ($t = 10$) to the Hebbian capacity at $T = 0$ is ≥ 3, refuted if the CI upper bound of the paired difference is < 0.1; **C2′** (shape) holds if the median curve $t \mapsto \hat{\alpha}_c$ is unimodal within 0.02 with an interior maximum t^*; **C3′** (control) holds if the CI upper bound of the ratio S of the shrink ($c(10)$) to the Hebbian capacity at $T = 0.2$ is < 1.1, refutes if the lower bound is ≥ 2; the $T = 0$ ratio must be exactly 1 on every seed, otherwise the run is void. **P4** (secondary label): spurious final states at $\alpha \in \{0.14, 0.20\}$ reduced by ≥ 50 %. **C4** (correlated run): paired difference $R(m = 0.30) - R(m = 0)$ with CI excluding 0. Seeds whose Hebbian $\hat{\alpha}_c$ is below the grid have an undefined ratio.
These are the confirmatory and correlated criteria. The pilot registration used the original C2 (monotonicity of the median curve, tolerated decrease 0.05 per grid step), C3 at $T = 0$ only, and P4 at loads 0.10 and 0.14; the confirmatory amendment replaced C2 by C2′ and moved C3 to $T = 0.2$ after the pilot had been observed, and the correlated registration was written after both independent-pattern studies. The pilot's median curve decreases from 0.600 to 0.522 between $t = 10$ and $t = 100$ and passed C2 only because no single step exceeded 0.05.

2.6 What was changed with respect to the original, and why

1. Finite $N \leq 1\,000$ and an empirical, noisy-cue, 0.5-crossing capacity estimator instead of the replica-theory $\alpha_c(t)$: absolute capacities are therefore not comparable

with the published curve; the replication tests ratios and qualitative claims (gain, spurious removal, projector limit).

2. Self-coupling removed ($J_{ii} = 0$) in all arms. The field printed in the original article sums over $j = 1 \dots N$ (their Eq. 2.16), and the later work of the original group and its students does not follow a single convention. Some algorithms remove the diagonal before the dynamics: "Remove the diagonal terms from J" opens Algorithm 1 of Agliari et al.[8], Algorithm 2 of Agliari et al.[6] as numbered in arXiv:2308.01421v2, where the kernel itself is computed with its diagonal (footnotes 4-5) and the diagonal is zeroed only for the dynamics, and Algorithms 4 and 7 of Aquaro's thesis[7]; a master's thesis supervised by Agliari sets $J_{ii} = 0$ and leaves it unchanged by unlearning[9] (Definition 5.1 and Eq. 6.2, pp. 35 and 52-56); the zero-temperature update of Agliari et al.[10] uses overlaps with site i excluded (Algorithm 1); and Barra et al.[11] subtract the diagonal contributions from their dense Hamiltonian (Eqs. 42-44). Others keep it: the 2022 preprint by the same six authors as[8] iterates $\sigma \leftarrow \text{sign}(J\sigma)$ with the full kernel[12] (Eqs. 2.41-2.42); the multidirectional dreaming model of Barra et al.[13] sums over all i, j (Eq. 4 and Algorithm 1 as numbered in arXiv:2605.13721v1); Alemanno's thesis[14] keeps it in the dreamed kernel (Eq. 2.29, p. 72) while excluding $j = i$ from the field of its reference Hopfield model (p. 45); and Agliari et al.[15] include self-interactions in their main analysis (Appendix E), set the diagonal to zero only in the training of their Appendix A, and note that in the Hopfield model the diagonal raises the one-step magnetisation from $\text{erf}(1/\sqrt{2\alpha})$ to $\text{erf}((1+\alpha)/\sqrt{2\alpha})$ (footnote 3). Table 1 lists fourteen further texts, from the same school and outside it. Two of them compare the two conventions, for the Hebbian kernel only. Folli, Leonetti and Ruocco[16] keep the diagonal and test whether each stored pattern is a fixed point after one parallel sign update: the diagonal adds P coherent terms to the local field ($N + P - 1$ instead of $N - 1$; Sec. 3.1 and footnote 2, p. 3), and the second region of near-perfect recall that they find at $P \gg N$ disappears when $J_{ii} = 0$ (Fig. 6 and text, p. 6). Vinci, Galluzzi and Mattia[17], for deterministic graded units with $J_{ii} = g\alpha$, find an overlap with the pattern used as initial state close to 1 at loads where the same networks with $J_{ii} = 0$ lose recall near $\alpha \approx 0.14$ (Fig. 3, p. 5), and critical temperatures of pure and mixture states that are lower when $J_{ii} = 0$ (Fig. 7, p. 9); they mention dreaming only as an alternative that their autapses make unnecessary (p. 10). None of these texts measures the effect of the convention on a dreamed-to-Hebb capacity ratio. NH-01 follows the first convention, which is also required for detailed balance under sequential updates (Proposition 1.3.1 of[7]). ReScience C asks that replication work "be based exclusively on the originally published paper, although exceptions are admitted if properly documented in the replication article"[18] (p. 4). The convention $J_{ii} = 0$ is such an exception, and we declare it as one: it is taken from the 2024-2025 algorithms of the original group cited above, not from the 2019 article, whose field (Eq. 2.16) keeps the diagonal. Its effect is measured by the post-freeze ablation (Table 5): at $t = 10$ the dream/Hebb ratio is 3.90 with the diagonal zeroed in both arms, 1.90 with it kept in both arms, and the mixed ratio (kept-diagonal dream over zero-diagonal Hebb) is 3.05. See also Benedetti et al.[19] (Sec. III C), for whom correcting the diagonal once at the end rather than at each step of an unlearning procedure changes which eigenvector is dreamed and lets the negative eigenvalue plateau overtake the positive one ("inversion"), and Krotov and Hopfield[20] (footnote 1), for whom excluding repeated indices from the energy of a dense associative memory changes the prefactor of its capacity. For ± 1 spins the diagonal only adds the configuration-independent constant $-\frac{1}{2}\sum_i J_{ii}$ to the energy; it acts through the zero-temperature update rule and hence through the capacity estimator. After the freeze of this package we pre-registered and ran an ablation keeping the diagonal (Section "Self-coupling convention", Table 5, Figure 3): the dream/Hebb ratio depends on the convention used in both arms (with the diagonal kept, neither confirmed nor refuted under rule C1, applied as a reading

aid), and the contribution of the self-coupling removal to the finite optimum in t is not isolated (registered test C5 inconclusive).

3. Dynamics and success criterion: asynchronous updates at $T = 0$ instead of the parallel sign update of the original zero-noise field and basin experiments (their Eq. 2.17); a fixed number of flipped cue bits, round$(0.10\,N)$, instead of independent flips with probability p; an absolute overlap threshold (0.9, or 0.8 at $T = 0.2$) instead of "largest Mattis overlap"; at $T = 0.2$, a fixed 40-sweep window without an equilibration diagnostic.
4. Matched shrink control and correlated patterns: additions, not changes.
5. Parameters and decision rules fixed before each study's main run (grids, seeds, thresholds); one voided launch of the pilot (BLAS thread oversubscription, killed after ~150 s before the main run; log kept in the upstream repository, D1).

Table 1. Self-coupling conventions in further texts (complement to item 2; pages as printed). "Kept": the sum over i, j includes $i = j$ in the energy or the field; "zero": $J_{ii} = 0$ is imposed. No text of this table measures the effect of the convention on a dreamed-to-Hebb capacity ratio.

Text	Convention	Where
Albanese et al. 2026[21], Hebbian BEG networks	kept, stated as "autapses"; the authors "speculate" that autapses contribute to the absence of a blackout catastrophe, citing[17]	Eqs. (2.2), (2.18), pp. 3, 6; Fig. 4 caption, p. 13; Fig. 7 discussion, p. 29
Albanese, Barra, Cirillo 2025[22], Gaussian couplings	zero, by definition	Definition 1, p. 3
Agliari, Duarte Mourão, Fachechi, Vivo 2026[23]	zero in the four matrices studied; the replica computation is done with the diagonal, then removed as a shift of the spectrum ($\bar{\lambda}(t)$ for the dreamed kernel, Lemma 1 and Theorem 1): a result on the limiting spectral density, not on fixed points	Eqs. (3), (5), (7), (8) and note 2, p-2; SM pp. p-13 to p-16
Agliari, Alessandrelli, Barra, Fachechi 2026[24]	kept, repeated indices included	Eqs. (2.2)-(2.4), p. 3
Agliari, Alessandrelli, Duarte Mourão, Fachechi 2025[25]	kept in the intra-layer field $J\sigma$, $J = N^{-1}\xi\xi^{\mathsf{T}}$	p. 2
Agliari, Lepre, Roscani 2026[26]	kept, prefactor $1/(2N)$	Eq. (2.2), p. 3
Agliari et al. 2026[27]	kept in the energy and in the field of the dynamics; Appendix H, on the dual of an isolated Hopfield network, writes $i < j$	Eq. (2.1), p. 2; Eq. (4.3), p. 4; p. 17
Personnaz, Guyon, Dreyfus 1985[28]	projection rule $C = AA^{I}$ with no zeroing of C_{ii} stated; the stability condition $C\sigma^{k} = \sigma^{k}$ is written for this full projector	Eq. (6), p. L-362; Eq. (7), p. L-363
Agliari, Barra, Bianco, Fachechi, Pallara 2026[29]	kept in the Hamiltonian (equilibrium theory)	Eq. (2.1), p. 4

Text	Convention	Where
Benedetti, Ventura, Marinari, Ruocco, Zamponi 2022[30]	zero: Hebbian $J_{ii} = 0$, reset to 0 after each unlearning step; "autapses" absent	Eq. (3), p. 1; Eq. (9), p. 3; p. 9
Alemanno 2026[31]	zero in the pairwise model; in the exponential model the self-coupling, "the exponential-model analogue of the diagonal term J_{ii}", is kept	Definition 2, p. 3; p. 6; p. 7
Barra 2008[32]	diagonal terms added to a cavity Hamiltonian give a contribution independent of the spins that vanishes after the disorder average (equilibrium, no dynamics)	Eq. (3.134), p. 97
Folli, Leonetti, Ruocco 2017[16]	kept, compared with zero for the Hebbian kernel (one-step stability of stored patterns)	pp. 3, 6
Vinci, Galluzzi, Mattia 2025[17]	kept ($J_{ii} = g\alpha$), compared with zero for the Hebbian kernel (graded units)	Fig. 3, p. 5; Fig. 7, p. 9

3 Results

3.1 Independent patterns

Pilot (D1), $N = 256$, $T = 0$, $t = 10$: Hebb $\hat{\alpha}_c = 0.175$ [0.167, 0.183] → dream 0.599 [0.591, 0.606], ratio **3.43 [3.26, 3.60]**, difference 0.423 [0.411, 0.435]; shrink ratio 1.000 on every seed, every N, every t. Ratios at $N = 64$ and 128: 3.13 [2.74, 3.57] and 3.38 [3.21, 3.53]. Confirmatory (D2), $N = 1\,000$, 10 seeds, 9 120 cells:

Criterion (cell)	Result	Label
C1 gain ($T = 0$, $t = 10$)	Hebb 0.157 [0.155, 0.160] → dream 0.607 [0.604, 0.610]; ratio **3.86 [3.79, 3.93]**; difference 0.450 [0.445, 0.454]	confirmed
C2′ shape ($T = 0$)	medians 0.158, 0.413, 0.571, 0.603, **0.616**, 0.606, 0.595, 0.582, 0.569, 0.527 for t = 0 … 100; $t^* = 7$ (9 seeds at 7, 1 at 10); drop $\hat{\alpha}_c(7) - \hat{\alpha}_c(100)$ = 0.088 [0.082, 0.092]	holds
C3′ control ($T = 0.2$, $t = 10$)	Hebb 0.126 [0.125, 0.128] → shrink 0.123 [0.121, 0.125]; ratio **0.975 [0.967, 0.982]**; $T = 0$ ratio exactly 1 on every seed and t	holds
P4 spurious ($T = 0$)	$\alpha = 0.14$: 0.159 → 0.000; $\alpha = 0.20$: 0.994 → 0.000	holds
Overall		confirmed

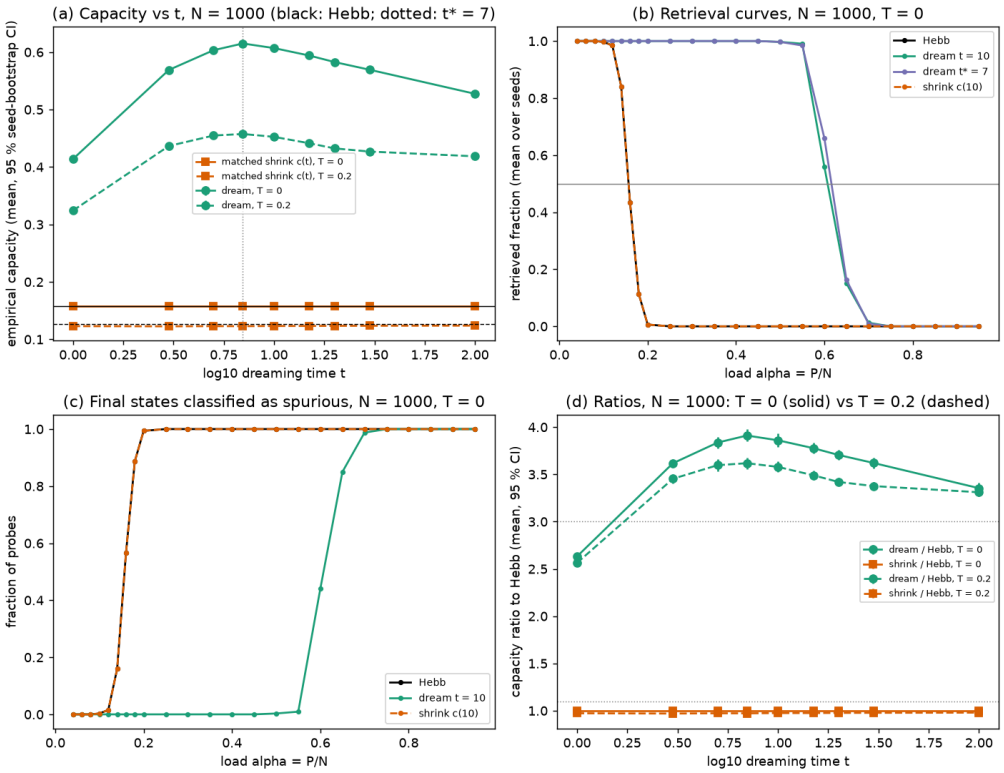


Figure 1. NH-01 confirmatory run, $N = 1\,000$, seeds 0–9 (D2; publication rendering by `code/scripts/publication_figures.py`, the runner's own figure is kept in `data/`). (a) Empirical capacity $\hat{\alpha}_c$ versus dreaming time t for the dreamed kernel (circles) and the matched uniform-shrink control (squares) at $T = 0$ (solid) and $T = 0.2$ (dashed); the horizontal lines are the Hebbian reference; the dotted vertical line marks $t^* = 7$. (b) Retrieved fraction versus load α at $T = 0$ for Hebb, dream $t = 10$, dream $t^* = 7$ and shrink $c(10)$. (c) Fraction of probes whose final state is classified as spurious, versus load, $T = 0$. (d) Capacity ratio to Hebb with seed-level 95 % bootstrap intervals; the dotted lines are the pre-registered thresholds 3.0 (confirmation) and 1.1 (control).

At $T = 0.2$ the dreaming gain persists: ratio 3.58 [3.53, 3.64] at $t = 10$, 3.62 [3.55, 3.68] at $t = 7$. At $T = 0$ the Hebbian estimate falls with size (0.211, 0.179, 0.175, 0.157 for N = 64, 128, 256, 1 000; the thermodynamic value is 0.138) while the dreamed capacity stays ≈ 0.60; the ratio grows for that reason. At loads 0.14 and 0.20 no dream $t = 10$ probe is classified as spurious; at higher loads some are (e.g. all 32 at $\alpha = 0.8$, seed 0), so the claim is load-specific.

3.2 Correlated patterns

Correlated run (D3), $N = 1\,000$, seeds 100–109, 29 640 cells, $t = 10$ unless stated.

Table 3. $T = 0$, $t = 10$ (C1, C2′).

Level m ($q = m^2$)	Hebb $\hat{\alpha}_c$	dream $\hat{\alpha}_c$	C1 ratio	C2′ t^* (per-seed range); drop to $t = 100$
0 (0)	0.160 [0.158, 0.162]	0.611 [0.606, 0.616]	**3.82 [3.77, 3.88]** — confirmed	7 (5–10); 0.084 [0.079, 0.089] — holds
0.15 (0.0225)	0.068 [0.066, 0.070]	0.616 [0.612, 0.619]	**9.06 [8.84, 9.29]** — confirmed	7 (7–10); 0.088 [0.085, 0.090] — holds
0.30 (0.09)	0.014 [0.014, 0.015] (every seed between loads 0.01 and 0.02)	0.636 [0.633, 0.639]	**44.38 [42.56, 46.97]** — registered C1 confirmed; crossings bracketed by loads 0.01–0.02, magnitude grid-sensitive (see text)	10 (7–10); 0.103 [0.098, 0.108] — holds

Table 4. $T = 0.2$, $t = 10$ (C3′) and level verdict.

Level m	Hebb $\hat{\alpha}_c$	C3′ shrink ratio	Level verdict
0	0.127 [0.125, 0.129]	**0.979 [0.974, 0.984]** — holds	**confirmed**
0.15	0.054 [0.053, 0.056]	**0.986 [0.979, 0.993]** — holds	**confirmed**
0.30	5/10 seeds below grid; zero-coded mean 0.007 [0.003, 0.011]	undefined on 5 seeds; zero-coded five-seed statistic 0.6 [0.2, 1.0], two of its shrink numerators also below grid	registered: confirmed; post-analysis assessment: **inconclusive**

Reading rule for $m = 0.30$. At $T = 0$ all ten Hebbian crossings are bracketed by the loads 0.01 and 0.02 (flags ok; the registration added these two loads for that purpose), so the registered ratio 44.38 is defined and its C1 label is confirmed. Its bootstrap interval describes seed variation conditional on the interpolation rule and does not include the load-grid discretisation error; reading the crossings as anywhere inside their bracketing intervals gives a ratio between about 30 and 65. We therefore do not interpret the magnitude as a precise model-level capacity ratio. At $T = 0.2$ five Hebbian estimates are below the grid; among the five seeds with a positive Hebbian denominator, two shrink estimates are also below the grid. The archived code replaces below-grid capacities by zero, so the Hebbian mean 0.007 [0.003, 0.011] and the five-seed shrink statistic 0.6 [0.2, 1.0] (per-seed 1, 0, 1, 1, 0) are outputs of that coding convention, not censoring-adjusted estimates. The absolute capacities at $T = 0$, also interpolated but less sensitive to the small denominator, are the more robust reading: the dreamed capacity barely moves with correlation (0.611, 0.616, 0.636) while the Hebbian capacity collapses (0.160, 0.068, 0.014),

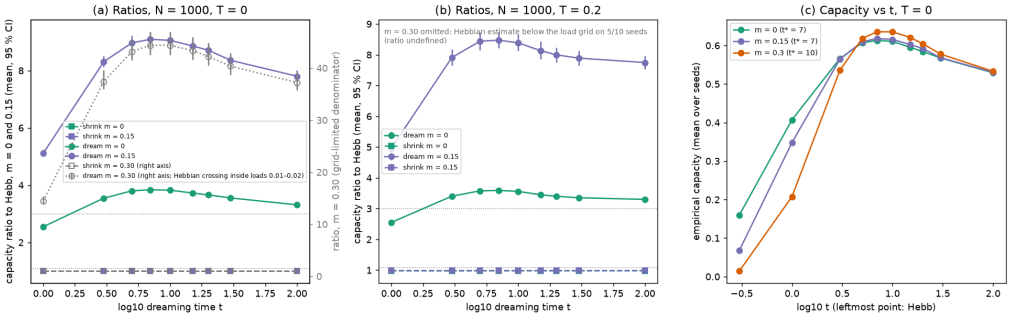


Figure 2. NH-01 correlated-pattern run, $N = 1\,000$, seeds 100–109 (D3; publication rendering by `code/scripts/publication_figures.py`). (a) Capacity ratio to Hebb versus dreaming time at $T = 0$ for ancestor overlap $m = 0$ and 0.15 (left axis; dream: circles, matched shrink: squares, 95 % seed-bootstrap intervals); the $m = 0.30$ curves are drawn in grey on the right axis because their Hebbian denominator is bracketed by the loads 0.01 and 0.02, so the ratio is defined but set by the load grid. (b) Same at $T = 0.2$; $m = 0.30$ is omitted because five of its ten Hebbian estimates, and two further shrink estimates, are below the grid, so its ratios are undefined or zero-coded. (c) Absolute capacity versus t at $T = 0$ for the three levels (leftmost point: Hebb), the robust reading at $m = 0.30$. The unmodified runner figure, whose automatic title labels every level "confirme", is kept in `data/`.

so the paired absolute gain grows 0.451 [0.445, 0.456] → 0.548 [0.543, 0.552] → 0.621 [0.618, 0.625]. The tagged code labels C3′ "holds" and the level "confirmed" in `summary.json`, computed on the five seeds with a positive denominator. This article keeps that registered label and adds, next to it, a post-analysis assessment of inconclusive for the level (decision of 2026-10-30, commit `5884927`): a methodological reservation, not a registered rule. Its ground is censoring: five denominators and two further numerators are below the grid and coded as zero, so the available bootstrap does not estimate the ten-seed control ratio; Hebbian retrieval at $T = 0.2$ survives only within the first load interval. The registered labels are kept unchanged in the raw outputs. The pre-registered below-grid rule was written for C1 only; a future amendment should extend it to C3′ and the load grid below 0.01.

C4 trend: the registered estimate is $R(0.30) - R(0)$ = 40.6 [38.7, 43.2]. The label is `gain_increases`. Its magnitude depends on the interpolation of the small Hebbian denominator at $m = 0.30$ and the interval does not account for load-grid resolution; we retain the increasing direction as a descriptive result only. P4 holds at every level (0 spurious final states under dreaming at $\alpha \in \{0.14, 0.20\}$); the $T = 0$ exactness check passed at every level. $m = 0$ reproduces the independent- pattern run on new seeds (3.82 vs 3.86; $t^* = 7$ in both; shrink 0.979 vs 0.975).

3.3 Internal reproduction of the $N = 1\,000$ studies

Both $N = 1\,000$ runs were re-executed on 2026-10-30 from a fresh clone checked out at the tag, with the SHA-256 of the executed files checked against `git show <tag>:<file>` before launch and the tag record fetched from the GitHub API before launch (D2 and D3, directory `provenance-rerun-20261030/`). Result: **9 120 / 9 120** and **29 640 / 29 640** cells identical to the committed `cells.jsonl.gz` on every field except the wall-clock `task_seconds` (cells matched by key, NaN compared as NaN); `summary.json` identical except its `run_meta` block; `figure.png` byte-identical (SHA-256 `3b64a94e…` and `de3ee720…`). The re-runs used 2 worker processes instead of 4 for the confirmatory run; the worker count changes only the scheduling.

3.4 Self-coupling convention (post-freeze ablation)

This subsection reports a study run after the submission package was frozen on 2026-10-31; every number above is unchanged. A fourth L1 pre-registration (S4: hypneumlab/nerve-wml, tag `prereg-nh01- diagonal-20261031`, 2026-10-31 16:38:17 UTC; run launched 58 s later; data D4) compares, at $N = 1\,000$, $T = 0$, on new seeds 200–209, two arms at every t of the confirmatory grid: `dream_offdiag`, an exact replica of the confirmatory dream arm ($J_{ii}(t) = 0$), and `dream_diag`, in which the diagonal of $J(t)$ is kept in the update field, as in the field printed in the original article. At $t = 0$ each arm is the Hebbian kernel in its own convention ($J_{ii} = P/N$ kept in `dream_diag`). Cues, dynamics, estimator and bootstrap are those of the confirmatory run; 4 800 cells, no non-converged probe, no right-censored estimate.

The registered question was the cause of the finite optimum $t^* = 7$, through the per-seed drop $D = \hat{\alpha}_c(7) - \hat{\alpha}_c(100)$. The positive control passed (D = 0.086 [0.082, 0.090] in `dream_offdiag`, overlapping the confirmatory 0.088 [0.082, 0.092]). **C5** (primary): D = 0.048 [0.043, 0.054] in `dream_diag`, **inconclusive** under the registered thresholds (≤ 0.020 for "cause", ≥ 0.050 with a lower bound above 0.020 for "not the cause"). **C6** (secondary): the paired difference of the drops is 0.038 [0.032, 0.043], positive on 10/10 seeds: removing the self-coupling contributes to the drop between $t = 7$ and $t = 100$, but whether it causes the finite optimum is not decided. With the diagonal kept the curve has another shape (Figure 3): it peaks on the grid at $t = 1$ (all ten seeds; the only grid point strictly between 0 and 3 is $t = 1$) and then decreases monotonically to 0.441 (median) at $t = 100$.

The headline ratio depends on the convention used in both arms (Table 5). The ablation did not register C1; its thresholds are applied here as a reading aid only.

Table 5. Dream/Hebb capacity ratio at $T = 0$, $t = 10$, by self-coupling convention. Mean over seeds, 95 % seed-level percentile bootstrap (10 000 resamples, seed 20261025); D2: confirmatory run, D4: ablation.

Convention (both arms)	Seeds	Hebb $\hat{\alpha}_c$	dream $\hat{\alpha}_c$, $t = 10$	Ratio	Reading under rule C1
$J_{ii} = 0$ (D2, this article)	0–9	0.157 [0.155, 0.160]	0.607 [0.604, 0.610]	**3.86 [3.79, 3.93]**	confirmed (registered)
$J_{ii} = 0$ (D4)	200–209	0.157 [0.155, 0.159]	0.611 [0.607, 0.614]	**3.90 [3.86, 3.94]**	above the C1 threshold (descriptive)
diagonal kept (D4)	200–209	0.251 [0.246, 0.256]	0.477 [0.475, 0.480]	**1.90 [1.86, 1.95]**	neither confirmed nor refuted under rule C1

Note to the table: a mixed ratio, the kept-diagonal dreamed capacity over the zero-diagonal Hebbian baseline, is 3.05 [3.00, 3.09]; it corresponds to no published protocol and is given for completeness only. Student-t intervals, as quoted in our comment on the submission issue, are [1.85, 1.96] for the kept-diagonal ratio and [2.99, 3.10] for the mixed one. For the kept-diagonal convention the lower bound 1.86 is below the confirmation threshold 3 and the paired difference, 0.226 [0.220, 0.233], is above the refutation bound 0.1.

The diagonal acts in opposite directions on the two arms. In the Hebbian baseline the self-coupling $J_{ii} = P/N$ adds an inertia term to every field and raises the noisy-cue estimate from 0.157 to 0.251 (at $\alpha = 0.20$, 298/320 probes retrieved against 3/320 with $J_{ii} = 0$); in the dreamed kernel at $t = 10$ it lowers the estimate from 0.611 to 0.477. At

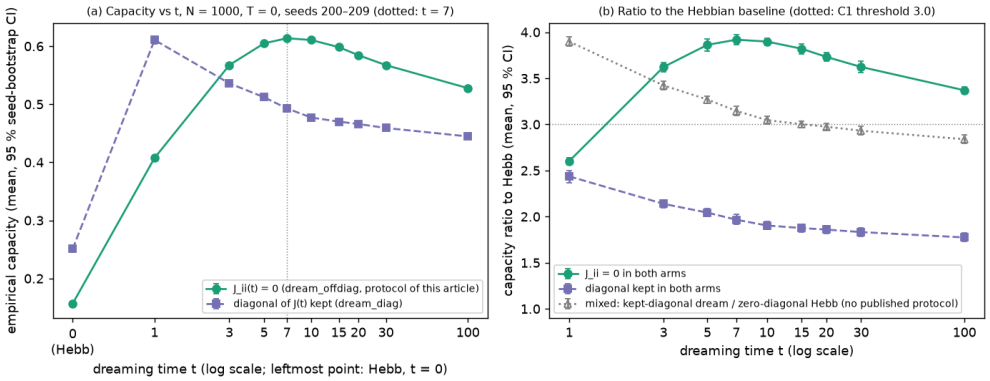


Figure 3. Post-freeze self-coupling ablation, $N = 1\,000$, $T = 0$, seeds 200–209 (D4; rendering by `code/scripts/figure_diagonal_ablation.py` from `summary.json`). (a) Empirical capacity versus dreaming time for the zero-diagonal arm (`dream_offdiag`, the protocol of this article, circles) and the kept-diagonal arm (`dream_diag`, squares); the leftmost point of each curve is the Hebbian kernel in the same convention; the dotted vertical line marks the registered $t^* = 7$. (b) Ratio to the Hebbian baseline, each arm over its own $t = 0$ point; grey: mixed ratio of the kept-diagonal dream to the zero-diagonal Hebbian baseline, which corresponds to no published protocol; the dotted line is the C1 threshold 3.0. Error bars: 95 % seed-level percentile bootstrap.

high load the kept-diagonal arm freezes near the cue: at $t = 7$, $\alpha = 0.50$ it retrieves 129/320 probes (318/320 with $J_{ii} = 0$) with a mean overlap of 0.90 with the cued pattern, against a retrieval threshold of 0.9 and a cue overlap of 0.80, so in that arm $\hat{\alpha}_c$ depends on the threshold more than in the zero-diagonal arm. The ablation does not test cue noise, synchronous dynamics, finite N, $T = 0.2$, correlated patterns or the matched-shrink control.

4 Conclusion

Under decision rules fixed before each study's main run, this partial replication **supports** the central qualitative claims of Fachechi, Agliari and Barra within our finite-size protocol: the dreamed kernel raises the empirical storage capacity of the Hopfield network by a large factor (3.4–3.9 at $t = 10$ for independent patterns, N = 256 to 1 000), leaves 0/320 probes classified as spurious at each of the loads 0.14 and 0.20 in the confirmatory run at $T = 0$, against 51/320 and 318/320 under Hebb (which does not establish the absence of spurious attractors elsewhere in state space), keeps its advantage under thermal noise, and behaves as the projector rule for large t (the pilot's $f = 0$ capacity estimate is right-censored at $\alpha = 0.95$ at the tested times $t \in \{10, 30, 100\}$ for $N \in \{64, 128, 256\}$ — a 50 %-retrieval criterion on at most 32 probed patterns, not a test that every stored pattern is a fixed point). The ratios 3.86 and 9.06 are results of this study, not numbers published in 2019. On correlated patterns the gain grows with correlation at $m \leq 0.15$, consistent with the projector limit's insensitivity to pattern correlations.

Two points go beyond, or qualify, the original. First, the added control shows that the Frobenius-matched uniform rescalings of the Hebbian couplings tested here do not reproduce the gain: they are inert at $T = 0$ by scale invariance and give a mean paired relative capacity reduction of 2.52 % at $T = 0.2$ (seed-bootstrap 95 % interval 1.80–3.35 %). The dreaming effect is a spectral change of the attractor landscape, not a global downscaling at equal synaptic budget. Second, with the self-coupling removed and a noisy cue, the finite-size capacity is not monotone in t: it peaks at $t^* = 7$ and loses 0.088 [0.082, 0.092] by $t = 100$. This does not contradict the mean-field $\alpha_c(t)$: our noisy-cue threshold differs from the equilibrium retrieval boundary of the theory. We observe a

lower empirical capacity at $t = 100$ than at the grid optimum $t = 7$ without measuring basin sizes or isolating the cause (self-coupling removal, cue noise, asynchronous dynamics, finite N). The original article does quantify retrieval basins, at $N = 1\,000$ and $\alpha = 0.05$ under parallel zero-noise dynamics, and finds them stabilised by dreaming; that experiment does not cover our higher-load protocol. An interior optimum in a dreaming budget is published elsewhere. For Hopfield-Feinstein-Palmer unlearning, Marinari, Rossi and Zamponi[33] report a recognition rate that peaks at a finite number of dreaming steps and then returns to zero, the optimal number growing with the load (Fig. 2a, p. 5), and an optimum in the balance of learning and dreaming steps that is a narrow line without synaptic bounds and a broad region with them (Figs. 4 and 5, pp. 6-7). In the multidirectional model of Barra et al.[13] the optimum is interior in the fraction of a fixed total dreaming budget given to one layer (Fig. 8, p. 13), while the retrieval region of the same article widens with t in its phase diagrams (Fig. 4, right, p. 10; Fig. 6, left, p. 12). Neither uses the closed-form kernel $J(t)$ with our estimator, and neither bears on the cause of the optimum observed here.

The t^* of this article is a dreaming time. It differs from the early-stopping time $t^*(t_d) \approx (2\bar{\lambda})^{-1} \log(1 + t_d\bar{\lambda})$, with $\bar{\lambda}$ the mean eigenvalue of the empirical correlation matrix of the stored vectors (Eq. 4.50 of Aquaro's thesis[7]), which is a training time. In the equivalence established by Agliari et al.[6] the dreamed kernel with dreaming time t_d is the ridge estimator with Tikhonov parameter $\epsilon_J = 1/t_d$, and $t^*(t_d)$ is the time at which unregularised gradient descent must be stopped to approximate it. Under that identity a finite optimal dreaming time reads as a finite optimal regularisation strength; this is a reading, not a result of this study. The same group reports that, when the stored vectors are noisy examples of archetypes, too long a dreaming time leads to overfitting (their Sections 4 and 5); that mechanism has no direct counterpart in our protocol, whose stored patterns are noise-free and whose noise is in the cue.

Recent work situates these observations. Benedetti et al.[19] follow the eigenvectors of the coupling matrix during unlearning, the spectral reading that distinguishes dreaming from a rescaling in our control. Serricchio et al.[34] combine reinforcement and removal in a "daydreaming" algorithm and report its effectiveness on correlated data generated by a random-features model, a correlation structure different from our single ancestor. Lucibello and Mézard[35] make precise, for dense associative memories, the distinction between the capacity of a typical pattern, of all patterns and the size of the basins — the distinction that also separates our noisy-cue estimator from a test of the stability of every stored pattern.

What this replication cannot settle: absolute capacities (finite-size, estimator-dependent), temperatures other than 0 and 0.2, correlation structures other than a single ancestor, and the $m = 0.30$ level, where the Hebbian crossings lie inside the first load interval (0.01–0.02) at $T = 0$ and below the grid on five seeds at $T = 0.2$. Nor does it settle whether dreaming removes mixture states as such: the `spurious` label of P4 is a residual class (neither retrieved nor another stored pattern), not a test of mixtures. Two tests would address this; neither was run. The first follows the stability of the symmetric 3-mixtures $\mathrm{sign}(\xi^1 + \xi^2 + \xi^3)$ as a function of t. The second is the spectral acceptance test of Agliari et al.[25]: a final state $\bar{\sigma}$ is accepted only if $\bar{\sigma}^\mathsf{T} J^K \bar{\sigma}/N > 0.8$, where J^K is the pseudo-inverse coupling matrix, obtained there as the fixed point of an iterative unlearning rule started from the Hebbian matrix; the stored patterns span the eigenspace of eigenvalue 1 of J^K, and spurious states fall outside it because of the nonlinearity of the sign (p. 3; Appendix C, p. 8). The threshold 0.8 is fixed there without calibration against a null. A scale sweep (nerve-wml#90) is open and not included here.

5 Reproducibility

The submission repository https://github.com/hypneum-lab/rescience-nh01 (MIT for the code, CC BY 4.0 for the text and figures) holds the runners (S1), the tests (S2), the three

pre-registrations (S3), the three data sets (D1–D3) and the provenance files of the internal re-runs. The runners are the files of `experiments/nh01_hopfield_dream/` of hypneum-lab/nerve-wml at the tag `prereg-nh01-correlated-20261027` (commit `df9a1e7`), whose SHA-256 are recorded in `code/CODE_SHA256_AT_TAG.txt`; the pilot tag freezes the pilot protocol only, the code it ran is the first runner commit after it. Software Heritage snapshot `swh:1:snp:01242c8b5873518a503b84a80ce3a302bb69b8a8` (visit 2026-10-29) holds the three tags as releases; the data are archived at Zenodo under DOI 10.5281/zenodo.23833431 (nerve-wml v1.10.0, concept DOI 10.5281/zenodo.19656342).
The post-freeze ablation adds its pre-registration (S4) and its data (D4, not in that Zenodo version). They were produced by the runner `diagonal_ablation.py` of hypneum-lab/nerve-wml at the tag of S4, which imports the frozen `run.py` unchanged (identical SHA-256).
Environment: Python 3.12.15, numpy 2.5.4, matplotlib 3.11.2 (full freeze in the re-run provenance directory of D2 and D3), single-threaded BLAS (the OpenMP, OpenBLAS, MKL and NumExpr thread counts set to 1), no GPU; host electron-server, 8 logical CPUs (Intel Xeon X5680, no AVX), Linux 6.8.0, shared with other jobs (load 7–20), `nice 19`. Every random draw comes from NumPy's default generator seeded by (20261025, seed, N, stream); bootstrap seed 20261025, 10 000 resamples.
Commands (from `code/`; wall times from each `run_meta.json`):

```
python -m experiments.nh01_hopfield_dream.run \
    --workers 4 --out results/nh01           # D1: 1 547 s
python -m experiments.nh01_hopfield_dream.confirmatory \
    --workers 4 --out results/nh01_confirmatory   # D2: 6 951 s
python -m experiments.nh01_hopfield_dream.correlated \
    --workers 2 --out results/nh01_correlated     # D3: 22 866 s
gzip -k results/<run>/cells.jsonl   # the comparator reads .gz
python -I scripts/compare_nh01.py \
    ../data/<run> results/<run> cmp.json
```

The confirmatory run started 14 s after its tag was pushed (2026-10-25, 13:47:22 UTC); the correlated run on 2026-10-27 at 16:48:06 UTC. The comparator does not set a failure exit status; it writes five fields into `cmp.json`:

```
keys_equal   cells_differing_non_timing   fields_differing
max_abs_diff   summary_equal_excl_run_meta
```

Exact reproduction means matching keys and zero non-timing differences; on another platform, the reported differences are read against the tolerance below. The exit status alone validates nothing. The option `--analyze-only` recomputes `summary.json` and `figure.png` from an existing `cells.jsonl`. Unit tests run in seconds; the pilot takes about 26 min on 8 cores; the two $N = 1\,000$ runs 2–8 h depending on the number of workers (the re-runs with 2 workers took 15 136 s and 30 547 s).
Expected outputs are the committed `summary.json` files. The two $N = 1\,000$ studies were reproduced on the original host with exact non-timing cell values and identical figure files; on another platform, cell keys and discrete outcomes (classifications, flags, sweeps) should match exactly and floating-point diagnostics (`c_fro`, `energy_own`) to about 10^{-12}, while PNG byte identity is not required. The pilot is not covered by the internal re-run.

6 Appendix: provenance table

Id	Content	Location in the submission repository
D1	pilot, $N \in \{64, 128, 256\}$, 28 080 cells	`data/nh01/` (`summary.json`, `cells.jsonl.gz`, `run_meta.json`, `figure.png`, `SHA256SUMS`)
D2	confirmatory, $N = 1\,000$, 9 120 cells, re-run provenance	`data/nh01_confirmatory/` (same files; `provenance-rerun-20261030/`)
D3	correlated, $N = 1\,000$, 29 640 cells, re-run provenance	`data/nh01_correlated/` (same files; `provenance-rerun-20261030/`)
D4	post-freeze self-coupling ablation, $N = 1\,000$, 4 800 cells (nerve-wml commit `d6c42c2`)	`data/nh01_diagonal_ablation/` (`summary.json`, `cells.jsonl.gz`, `run_meta.json`, `run.log`, `launch_record.md`, `SHA256SUMS`)
S1	runners `run.py`, `confirmatory.py`, `correlated.py` at the correlated tag	`code/experiments/nh01_hopfield_dream/`; hashes in `code/CODE_SHA256_AT_TAG.txt`
S2	unit tests (20 functions, 23 cases)	`code/tests/test_nh01_hopfield_dream.py`
S3	the three pre-registrations, verbatim copies of the tagged files (tag names in `prereg/README`)	`prereg/`
S4	post-freeze pre-registration of the self-coupling ablation, verbatim copy at tag `prereg-nh01-diagonal-20261031` (revision `swh:1:rev:6e1b006…`)	`prereg/2026-10-31-nh01-diagonal-ablation.md`
F	publication rendering of the figures	`code/scripts/publication_figures.py`; `code/scripts/figure_diagonal_ablation.py` (D4)

7 AI usage

Claude Code (Anthropic) was used as a tool for the code, the documentation and drafts of this text, in sessions driven by the author; every draft was read, corrected and arbitrated by the author. Codex CLI (OpenAI; gpt-6.1-sol, then gpt-6-astra) ran two automated checks of the finished package before submission; the briefs, reports and verdict tables are in `review/`. These checks are not peer review. The design, the three pre-registrations, every verdict and every number in this article were checked and decided by the author. Development history: hypneum-lab/nerve-wml pull requests #81, #86 and #87, issue #68.

References

1. A. Fachechi, E. Agliari, and A. Barra. "Dreaming neural networks: forgetting spurious memories and reinforcing pure ones." In: **Neural Networks** 112 (2019), pp. 24–40. DOI: 10.1016/j.neunet.2019.01.006. arXiv:1810.12217.
2. J. J. Hopfield, D. I. Feinstein, and R. G. Palmer. ""Unlearning" has a stabilizing effect in collective memories." In: **Nature** 304 (1983), pp. 158–159. DOI: 10.1038/304158a0.
3. I. Kanter and H. Sompolinsky. "Associative recall of memory without errors." In: **Physical Review A** 35 (1987), pp. 380–392. DOI: 10.1103/PhysRevA.35.380.
4. D. J. Amit, H. Gutfreund, and H. Sompolinsky. "Storing infinite numbers of patterns in a spin-glass model of neural networks." In: **Physical Review Letters** 55 (1985), pp. 1530–1533. DOI: 10.1103/PhysRevLett.55.1530.
5. G. Tononi and C. Cirelli. "Sleep and the price of plasticity: from synaptic and cellular homeostasis to memory consolidation and integration." In: **Neuron** 81.1 (2014), pp. 12–34. DOI: 10.1016/j.neuron.2013.12.025.
6. E. Agliari, F. Alemanno, M. Aquaro, and A. Fachechi. "Regularization, early-stopping and dreaming: a Hopfield-like setup to address generalization and overfitting." In: **Neural Networks** 177 (2024), p. 106389. DOI: 10.1016/j.neunet.2024.106389. arXiv:2308.01421.
7. M. Aquaro. "Applications of Spin Glass Theory in Neural Networks and Evolutionary Models." PhD thesis. Sapienza Università di Roma, Department of Mathematics "Guido Castelnuovo", 2025. URL: https://hdl.handle.net/11573/1739810.
8. E. Agliari, F. Alemanno, M. Aquaro, A. Barra, F. Durante, and I. Kanter. "Hebbian dreaming for small datasets." In: **Neural Networks** 173 (2024), p. 106174. DOI: 10.1016/j.neunet.2024.106174.
9. A. Canali. "Intelligenza artificiale nella fase REM." Supervisor E. Agliari; in Italian. MA thesis. Sapienza Università di Roma, Department of Mathematics "Guido Castelnuovo", 2017. URL: https://www.adrianobarra.com/uploads/3/7/8/8/37889083/alessandro_canali_tesimagistralematematica.pdf.
10. E. Agliari, A. Barra, A. Ladiana, and A. Lepre. **Exponential capacity in multilayer hetero-associative neural networks**. 2026. arXiv:2607.29554.
11. A. Barra, M. S. Centonze, M. Marra Solazzo, and D. Tantari. **On the supra-linear storage in dense networks of grid and place cells**. 2025. arXiv:2511.02441.
12. M. Aquaro, F. Alemanno, I. Kanter, F. Durante, E. Agliari, and A. Barra. **Recurrent neural networks that generalize from examples and optimize by dreaming**. 2022. arXiv:2204.07954.
13. A. Barra, F. Durante, A. Ladiana, and M. Marra Solazzo. "Do Hopfield networks dream of stored patterns? A statistical–mechanical theory of dreaming in multidirectional associative memories." In: **Neural Networks** 206 (2027), p. 109729. DOI: 10.1016/j.neunet.2026.109729. arXiv:2605.13721.
14. F. Alemanno. "Neural Networks and Learning Machines: from mathematical foundation to applications in biological complexity." PhD thesis. Università del Salento, Department of Mathematics and Physics "Ennio De Giorgi", 2021. URL: https://www.adrianobarra.com/uploads/3/7/8/8/37889083/alemanno_phdthesis.pdf.
15. E. Agliari, A. Fachechi, and D. Luongo. **The importance of being empty: a spectral approach to Hopfield neural networks with diluted examples**. 2025. arXiv:2503.15353.
16. V. Folli, M. Leonetti, and G. Ruocco. "On the maximum storage capacity of the Hopfield model." In: **Frontiers in Computational Neuroscience** 10 (2017), p. 144. DOI: 10.3389/fncom.2016.00144.
17. G. V. Vinci, A. Galluzzi, and M. Mattia. **Beyond catastrophic forgetting in associative networks with self-interactions**. 2025. arXiv:2504.04560.
18. N. P. Rougier, K. Hinsen, F. Alexandre, T. Arildsen, L. A. Barba, F. C. Y. Benureau, C. T. Brown, P. de Buyl, et al. "Sustainable computational science: the ReScience initiative." In: **PeerJ Computer Science** 3 (2017), e142. DOI: 10.7717/peerj-cs.142. arXiv:1707.04393.
19. M. Benedetti, L. Carillo, E. Marinari, and M. Mézard. "Eigenvector dreaming." In: **Journal of Statistical Mechanics: Theory and Experiment** 2024 (2024), p. 013302. DOI: 10.1088/1742-5468/ad138e.
20. D. Krotov and J. J. Hopfield. "Dense associative memory for pattern recognition." In: **Advances in Neural Information Processing Systems 29**. 2016, pp. 1172–1180. arXiv:1606.01164.
21. L. Albanese, A. Alessandrelli, A. Barra, and E. N. M. Cirillo. **Serial vs parallel recall in the Blume-Every-Griffiths neural networks**. Title as printed and as returned by the arXiv API; the model is called Blume-Emery-Griffiths in the text. 2026. arXiv:2601.07777.
22. L. Albanese, A. Barra, and E. N. M. Cirillo. "Guerra interpolation for inverse freezing." In: **Journal of Mathematical Physics** 66.12 (2025). Pages cited from arXiv v1, p. 123305. DOI: 10.1063/5.0280543. arXiv:2505.06202.
23. E. Agliari, P. Duarte Mourão, A. Fachechi, and P. Vivo. **Spectral criteria for generalization in unsupervised Hebbian nets**. 2026. arXiv:2606.04651.
24. E. Agliari, A. Alessandrelli, A. Barra, and A. Fachechi. **Dense auto-hetero associative memories applied to noisy communication channels**. 2026. arXiv:2609.32605.
25. E. Agliari, A. Alessandrelli, P. Duarte Mourão, and A. Fachechi. **Multi-channel pattern reconstruction through L-directional associative memories**. Pages cited from arXiv v2. 2025. arXiv:2503.06274.

26. E. Agliari, A. Lepre, and E. Roscani. **Classifications in modular restricted Boltzmann machines**. 2026. arXiv:2610.08612.
27. E. Agliari, A. Alessandrelli, A. Barra, M. Centonze, and F. Ricci-Tersenghi. "Networks of Hebbian networks: more is different." In: **Neural Networks** 194 (2026). arXiv title: Networks of neural networks: more is different, p. 108181. DOI: 10.1016/j.neunet.2025.108181. arXiv:2501.16789.
28. L. Personnaz, I. Guyon, and G. Dreyfus. "Information storage and retrieval in spin-glass like neural networks." In: **Journal de Physique Lettres** 46.8 (1985), pp. L359–L365. DOI: 10.1051/jphyslet:01985004608035900.
29. E. Agliari, A. Barra, P. Bianco, A. Fachechi, and D. Pallara. "The infinite volume limit of the Hopfield neural networks in the replica-symmetric regime." In: **Annales Henri Poincaré** (2026). Published online 2026-10-14; pages cited from arXiv v2. DOI: 10.1007/s00023-026-01770-4. arXiv:2409.10145.
30. M. Benedetti, E. Ventura, E. Marinari, G. Ruocco, and F. Zamponi. "Supervised perceptron learning vs unsupervised Hebbian unlearning: approaching optimal memory retrieval in Hopfield-like networks." In: **The Journal of Chemical Physics** 156.10 (2022). Pages cited from the arXiv version, p. 104107. DOI: 10.1063/5.0084219. arXiv:2201.00032.
31. F. Alemanno. **Attention over noisy examples: generalization and memorization in the exponential Hopfield network**. Pages cited from arXiv v2. 2026. arXiv:2610.22191.
32. A. Barra. "Mean field spin glasses: exact results without the replica trick." PhD thesis. 2008. URL: https://www.adrianobarra.com/uploads/3/7/8/8/37889083/tesidottoratobarra.pdf.
33. E. Marinari, S. Rossi, and F. Zamponi. **Dreaming improves memorization in a Hopfield model with bounded synaptic strength**. 2026. arXiv:2603.09384.
34. L. Serricchio, D. Bocchi, C. Chilin, R. Marino, M. Negri, C. Cammarota, and F. Ricci-Tersenghi. "Daydreaming Hopfield networks and their surprising effectiveness on correlated data." In: **Neural Networks** 186 (2025), p. 107216. DOI: 10.1016/j.neunet.2025.107216.
35. C. Lucibello and M. Mézard. "Exponential capacity of dense associative memories." In: **Physical Review Letters** 132 (2024), p. 077301. DOI: 10.1103/PhysRevLett.132.077301.